ORB -> VWAP Fade — Strategy v1.1

Validated specification + application guide + full PowerLanguage code

Concept

Auction Market Theory, responsive activity: when the open pushes price to an extreme of the first-N-minute range, fade that extension back to value (session VWAP). Works on ES because ES mean-reverts intraday. The edge lives in the reversion distance, which scales with how wide the opening range is — so the strategy only trades wide-open (high-volatility) days. Reverse-engineered from kusho's ORB / AMT idea plus an opening-range-width filter derived from the data (not hand-fitted).

Rules

- Instrument / data: **@ES, 1-minute bars**, RTH session, **SessStart = 930**.
- Opening range = first **7 minutes** of the session.
- Width gate: trade only if OR width ≥ **0.15%** of price.
- Entry (limit orders, so near-zero slippage): price tags the OR **high** -> **short**; price tags the OR **low** -> **long**.
- Take profit = **session VWAP** (volume-weighted, resets daily).
- Stop = **1.25 × OR range** beyond the faded extreme.
- Flat at end of day, **one trade per day**, both directions.

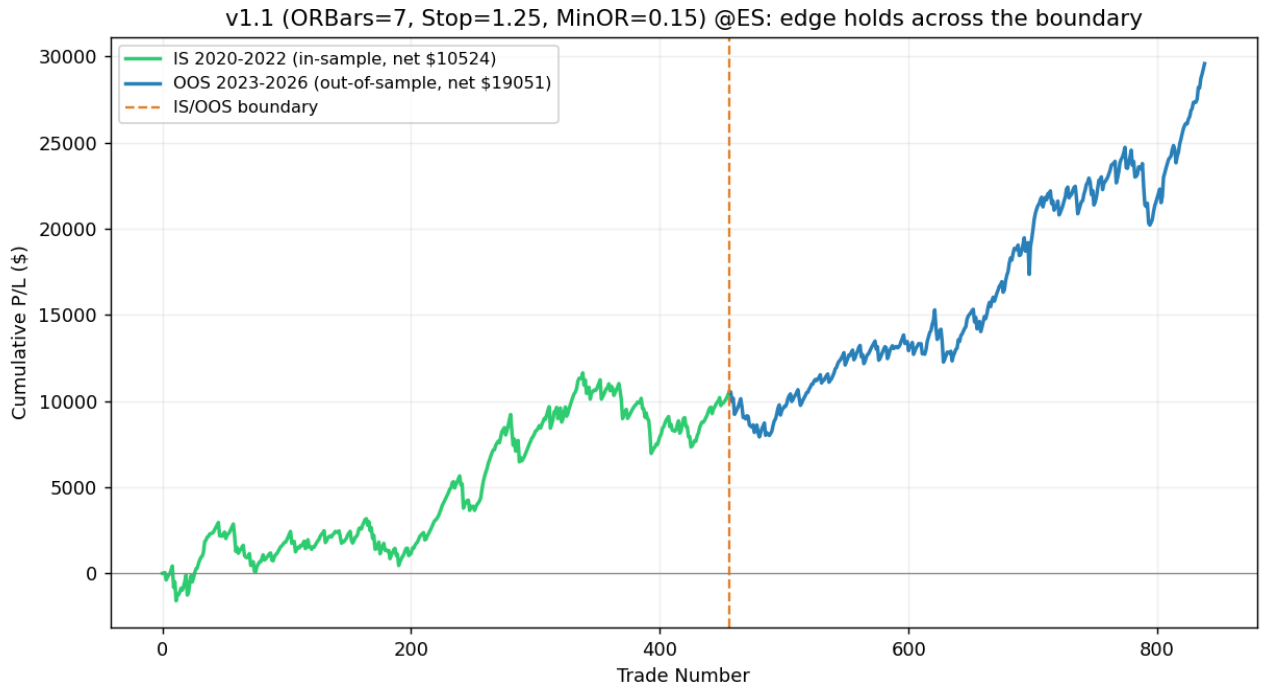
Parameters

Input	Value
SessStart	930
ORBars	7
StopMult	1.25
MinORpct	0.15
Costs	$1.5/side + 1 tick

Input	Value
MaxTradesPerDay	1
EODExit / SessEnd	true / 1545
UseWidthFilter	true
MaxORpct	100

Results (1 ES contract, with costs)

	IS 2020-2022	OOS 2023-2026	Full 2020-2026
Net profit	$10,524	$19,051	~$29,575
Win rate	76.3%	75.7%	~76%
Avg trade	$23.1	$49.9	~$36
Profit factor	1.23	1.43	~1.30
Max drawdown	-$5,074	-$5,101	-$5,100
RoMaxDD	2.07	3.74	~5.7



Equity curve: the edge continues seamlessly from in-sample (green) to out-of-sample (blue).

Robustness

- **Walk-forward:** win rate and drawdown almost identical IS vs OOS (76.3%->75.7%, -$5.07k->-$5.10k). OOS did not decay — the opposite of overfitting.
- **Parameter plateau:** ORBars 7 / Stop 1.25 / MinOR 0.15 sits at the center of a broad plateau (neighborhood median RoMaxDD ~4.5, 100% of neighbors profitable) — chosen as a robust center, not a peak.
- **Monte Carlo (10k resamples):** ~97% of histories profitable; max consecutive losses of ~6 is normal (up to ~11 possible), so no 'stop after X losses' rule is needed. Plan risk for the ~$8-9k 95th-percentile drawdown, not the ~$5k observed.

How to apply (MultiCharts)

1. Open the PowerLanguage Editor, create a new **Signal**, paste the code on the next page, compile.
2. Add the signal to an **@ES chart, 1-minute compression**, RTH session.
3. In Strategy Properties set **Commission = $1.5/side** and **Slippage = 1 tick**.
4. Confirm inputs: SessStart=930, ORBars=7, StopMult=1.25, MinORpct=0.15, UseWidthFilter=true.
5. Verify SessStart is 930 (NOT 830) — 830 breaks the strategy on this feed; confirm the feed timezone.
6. Size positions against the ~$8-9k Monte-Carlo drawdown, not the observed ~$5k.

Notes & caveats

- OOS (2023-2026) was a higher-volatility regime, which lifted the average trade ($50 vs $23). Don't extrapolate $50/trade; the trustworthy numbers are the ~76% win rate and the controlled drawdown.
- Entries and take-profits are limit orders, so realized slippage is near zero; only the stop-outs pay slippage.
- Provenance: reverse-engineered step by step from kusho's public ORB / AMT posts (ES, 1-min, VWAP target, both directions) plus a data-derived opening-range-width filter.

PowerLanguage code (v1.1, confirmed)

Insert as a Signal. Defaults below are the validated parameters.

```
[LegacyColorValue = true]

{ ================================================================
  ORB -> VWAP Fade  v1.1  (CONFIRMED / VALIDATED)
  MultiCharts PowerLanguage -> SIGNAL.  >>> @ES 1-MINUTE <<<
  ----------------------------------------------------------------
  This is the validated variant. Defaults below are the parameters
  that passed walk-forward IS(2020-2022)/OOS(2023-2026):
      SessStart 930 | ORBars 7 | StopMult 1.25 | MinORpct 0.15
  Results (1 ES contract, $1.5/side comm + 1 tick slip):
      Full  : Net ~$29.5k , PF ~1.30 , WinRate ~76% , MaxDD ~-$5.1k
      IS    : WinRate 76.3% , MaxDD -$5.07k
      OOS   : WinRate 75.7% , MaxDD -$5.10k   (edge held, no decay)
  Plan risk for the ~$8-9k Monte-Carlo 95th-pct drawdown, not the $5k.
  ----------------------------------------------------------------
  Logic: fade the opening-range extreme back to session VWAP, only on
  wide-open days (OR width >= MinORpct % of price). Both directions,
  one trade/day, stop beyond the range, flat at EOD.
  NOTE: SessStart=930 is validated on this dxFeed ES data. 830 breaks
  it -- confirm the feed timezone before changing it.
  ================================================================ }

Inputs:
    SessStart(930),
    ORBars(7),              { OR duration in 1-min bars }
    StopMult(1.25),         { stop = StopMult * OR range beyond the faded extreme }
    EntryCutoff(1500),      { inert in practice; kept for completeness }
    EODExit(true),
    SessEnd(1545),
    MaxTradesPerDay(1),
    UseWidthFilter(true),
    MinORpct(0.15),         { trade only if OR width >= this % of price }
    MaxORpct(100.0),
    LogDiag(false),
    DiagPath("C:\Temp\orb_diag.csv");

Variables:
    SessBar(0),
    ORHigh(0), ORLow(0), ORRange(0), ORwPct(0), ORLocked(false),
    LongStp(0), ShortStp(0),
    WidthOK(true), DirTxt(""),
    InSess(false),
    TypP(0), CumPV(0), CumV(0), VWAP(0);

InSess = (Time > SessStart) and (Time <= SessEnd);

if Date <> Date[1] then begin
    SessBar = 0;
    ORLocked = false;
end;

if InSess then SessBar = SessBar + 1;
```

PowerLanguage code (continued)

```
{ ---- session VWAP (reset at first RTH bar, volume-weighted) ---- }
if InSess and SessBar = 1 then begin
    CumPV = 0;
    CumV  = 0;
end;

if InSess then begin
    TypP  = (High + Low + Close) / 3;
    CumPV = CumPV + TypP * Volume;
    CumV  = CumV + Volume;
    if CumV > 0 then VWAP = CumPV / CumV;
end;

{ ---- build the opening range ---- }
if InSess and SessBar = 1 then begin
    ORHigh = High;
    ORLow  = Low;
end;

if InSess and SessBar > 1 and SessBar <= ORBars then begin
    if High > ORHigh then ORHigh = High;
    if Low  < ORLow  then ORLow  = Low;
end;

{ ---- LOCK: freeze range, width, stops ---- }
if InSess and SessBar = ORBars and not ORLocked then begin
    ORLocked = true;
    ORRange  = ORHigh - ORLow;
    if Close <> 0 then ORwPct = ORRange / Close * 100 else ORwPct = 0;

    LongStp  = ORLow  - StopMult * ORRange;
    ShortStp = ORHigh + StopMult * ORRange;

    WidthOK = (not UseWidthFilter) or (ORwPct >= MinORpct and ORwPct <= MaxORpct);

    if LogDiag then
        Print(File(DiagPath),
            NumToStr(Date, 0), ",",
            NumToStr(ORwPct, 4), ",",
            NumToStr(ORRange, 2), ",",
            NumToStr(Close, 2));
end;

{ ---- entries: fade the extreme back to VWAP, one per day ---- }
if ORLocked and WidthOK and InSess and (Time <= EntryCutoff)
   and (EntriesToday(Date) < MaxTradesPerDay) and MarketPosition = 0 then begin
    if ORLow  < VWAP then Buy("Fade-L")       next bar at ORLow  limit;
    if ORHigh > VWAP then SellShort("Fade-S") next bar at ORHigh limit;
end;

{ ---- exits: TP = live VWAP ; protective stop = frozen level ---- }
```

PowerLanguage code (continued)

```
if MarketPosition = 1 then begin
    Sell("L-VWAP") next bar at VWAP limit;
    Sell("L-Stp")  next bar at LongStp stop;
end;

if MarketPosition = -1 then begin
    BuyToCover("S-VWAP") next bar at VWAP limit;
    BuyToCover("S-Stp")  next bar at ShortStp stop;
end;

{ ---- end-of-day flat ---- }
if EODExit and (Time >= SessEnd) and MarketPosition <> 0 then begin
    Sell("EOD-L") next bar at market;
    BuyToCover("EOD-S") next bar at market;
end;
```